

Syllabus (2026)

Site: elearning.unideb.hu
Course: High-level programming languages 1 / Lab (INBPA0211L),
2025/26/2
Book: Syllabus (2026)

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Description

Syllabus for High-level Programming Languages 1

INBPA0211-21

Semester: 2

Type: Lecture / Laboratory

Number of Classes: 2+0+2

Credit: 6

Status: Obligatory

Assessment: Exam

Prerequisites: INBPA0104-21 (Introduction to programming)

Responsible: Dr. László Szathmáry

Contains information only for the lab part. Must be considered as a draft until the end of the first study week.

Table of contents

- 1. Overview**
- 2. Weekly Schedule**
- 3. Ethical and technical guidelines**
- 4. Requirements and examination**
- 5. Test specification**
- 6. Contact information**

1. Overview

By registering for the course in Neptun, you automatically accept the rules and regulations detailed on the following pages.

The same rules apply to all students, no exceptions are made.

Topics

The main goal of this course is to introduce basic programming concepts and tools with the use of a programming language that supports imperative programming. By the end of the semester, the students will be able to plan and implement simple programs, to read complex source codes and to proceed into an advanced course of programming.

The course covers the following basic programming tools. Compiler and interpreter. Variables, constants. Local and global variables. Scopes, lifetime. Control statements. Types, operators, operands. I/O tools. Functions, evaluation and pass-by techniques. Recursion, function call. Using the memory. Error handling; syntax and semantic errors. Command line arguments. Using libraries. Some thoughts will be given about basic computer science topics like Neumann-architecture, number systems, programming paradigms and history of programming languages.

Compulsory Reading:

- Robert W. Sebesta: Concepts of Programming Languages Addison-Wesley, 2015

Recommended Readings:

- Brian W. Kernighan, Dennis M. Ritchie: The C Programming Language (2nd ed.). Englewood Cliffs, NJ: Prentice Hall, 1988
- Bjarne Stroustrup: The C++ Programming Language (4th edition), Addison-Wesley, 2013

2. Weekly Schedule

Structure of the labs

According to the content of the further section *Discussion*, each lab (except testing) has the following structure:

1. *Warm-up Q&A session*: Decitated minutes for discussing the online received *restrospective professional and technical questions* (if they exist).
2. General content: the instructor explains the new topic(s); students code together with the instructor or manage their individual tasks (may vary).
3. *Cool-down Q&A session*: At most, ten (10) dedicated minutes are spent asking public interest questions about the current session's content.

Note that each question should be public interest. Moreover, they should meet the *Ethical and technical guidelines* (published in Moodle).

Schedule of the labs

All the groups will discuss the following topics during the semester. Note that national holidays and other events, such as the Professional Days, can affect this schedule. Thus, it is not guaranteed that different groups will be synchronized and minor changes can be made. Check the *Weekly schedule* section of your lab for the accurate schedule.

1. **Topic A1**: The environment (`gcc` compiler, VS Code, Hello World), compilation vs. runtime.
2. **Topic A2**: Language basics #1 (types, variables, expressions, statements, handling standard I/O, built-in functions, procedures, recursion).
3. **Topic A3**: Language basics #2 (types, variables, expressions, statements, handling standard I/O, built-in functions, procedures, recursion).
4. **Topic A4**: Language basics #3 (types, variables, expressions, statements, handling standard I/O, built-in functions, procedures, recursion).
5. **Topic B1**: Data structures and built-in functions #1 (pointers, 1D numeric arrays, 1D character arrays).
6. **Topic B2**: Data structures and built-in functions #2 (strings, dynamic memory allocation).
7. **Topic B3**: Data structures and built-in functions #3 (multidimensional arrays, consuming command-line arguments).
8. **Topic C1**: Data handling #1 (structures, reading and writing files, processing CSV documents, using functions `qsort()` and `bsearch()`, representing data structures).
9. **Topic C2**: Data handling #2 (structures, reading and writing files, processing CSV documents, using functions `qsort()` and `bsearch()`, representing data structures).
10. **Topic C3**: Data handling #3 (structures, reading and writing files, processing CSV documents, using functions `qsort()` and `bsearch()`, representing data structures).
11. **Topic X**: Warmup session.
12. **Lab test**: The primary event to get the signatures.
13. **Lab test (repeat occasion)**: An additional event to get the signatures.

Weekly schedule for Monday labs

We expect the following schedule for the labs on this day:

1. **02-09**: Topic A1
2. **02-16**: Topic A2
3. **02-23**: Topic A3
4. **03-02**: Topic A4
5. **03-09**: Topic B1
6. **03-16**: Topic B2
7. **03-23**: Topic B3
8. **03-30**: No lab (*Spring break*)
9. **04-06**: No lab (*Holiday*)
10. **04-13**: Topic C1
11. **04-20**: Topic C2
12. **04-27**: Topic C3
13. **05-04**: Lab test
14. **05-11**: Lab test (repeat occasion)

Weekly schedule for Wednesday labs

We expect the following schedule for the labs on this day:

1. **02-11**: Topic A1
2. **02-18**: Topic A2
3. **02-25**: Topic A3
4. **03-04**: Topic A4
5. **03-11**: Topic B1
6. **03-18**: Topic B2

7. **03-25:** Topic B3
8. **04-01:** *No lab (Spring break)*
9. **04-08:** Topic C1
10. **04-15:** Topic C2
11. **04-22:** Topic C3
12. **04-29:** Topic X
13. **05-05: Lab test**
14. **05-12: Lab test (repeat occasion)**

3. Ethical and technical guidelines

By registering for the course in Neptun, you automatically accept the rules and regulations detailed below.

The same rules apply to all students, no exceptions are made.

In the semester students must rigorously follow the basic ethical standards and the corresponding regulations of the University. Based on the previous semesters' experience, we would like to introduce a guide containing the most crucial examples:

1. The **official language of this course is English**. Thus, we provide only English materials and students should understand them. Students cannot use Google Translate or similar applications to translate our instructions. Students can communicate just and only in English during the labs, tests, and consultations.
2. **Neptun related issues are out of the instructors responsibility.**
Instructors do not have an administrative role in Neptun.
We cannot help you to resolve problems like:
 - registration issues,
 - group conflicts or changing them,
 - remaining in or moving from the collection group,
 - exam registration issues,
 - late payment for retake exam. (advised to hold a small amount in the payment account inside Neptun to cover fees)
3. A **well-behaved attitude** is expected in the labs and lectures.
Pay particular attention to the tone used for communication. Proclamatory style, jeer, and increased tone are not acceptable and should be avoided.
Disciplinary proceedings (investigations with penalties) are automatically initiated in these and similar cases.
4. In order to provide **equal conditions and a calm environment** for all the students, oral communication is forbidden during the tests. Instructors cannot answer questions related to the exercises. Instructors cannot verify solutions or help with technical details. An exception is that a student discovers a typo in the exercise sheet or our provided codes. In these cases, instructors fix the detected mistakes ASAP and may reward the student based on their discrete decision.
5. **All the cheating actions are strictly forbidden.**
Especially: Using switched-on electronic devices, such as mobile phones, wireless headphones, smart watches, etc. is forbidden. According to the previous statement, it is forbidden to wear or use any watch with a digital display. Note that a digital display clock with an analog clock face is still a digital display clock, thus, it is forbidden. The instructor may ask you any time to show the watch you wear to him or her. On the exams, these devices should be placed in your bag or jacket. Only your ID card and pen are allowed to be kept on the desk; as a consequence, other belongings cannot be stored on the desk. Students can only touch the permitted tools. *Please check further limitations in the University's ethical and examination rules!*
6. Any form of **communication is forbidden during the tests and exams** and automatically terminates both parties' examination with a failed grade.
7. **Recording or taking pictures of the labs and the lectures is forbidden.**
8. **Delays are not accepted during the tests and exams.** A student cannot start it if he/she arrives later than the official start time.
9. **Students must read and be aware of all the official announcements.** We use a Moodle course to support the work during the semester. *The student has the responsibility to check the content of the Moodle course, process, and be aware of all the published information. The same applies to the Neptun messages.*
10. **A Q&A session (a Moodle forum) will be available for each topic.** Students can post their questions using these activities. Instructors check the questions posted before the last but one workday before the next lab and reflect on them in the upcoming lab/lecture.
11. **Instructors** will strictly follow the guidelines; thus, they **cannot consider any inquiries or make any offerings**. Moreover, these attempts (both personally or via email) will be considered as unethical actions.
12. **Follow the dress code:** proper clothing should followed on the examinations.
Wearing the following parts is NOT allowed during the tests and exams: *hats and hoodies on the head, winter jackets, sun and smart glasses, ...*
Ears and wrists need to be well visible to avoid misconceptions.

Lab-related rules:

1. **Students must switch off their mobile phones and disconnect all wireless devices. Students cannot touch these devices while the laboratory computer's screen is unlocked.** During the lab test, these devices are placed under the PC's monitor, having the screen downside.*
2. Students can use their **own keyboards** during the lab tests. However, their **setup and removal are the student's own responsibility**. Instructors cannot give extra time or help during these actions. Moreover, instructors are not responsible for any negative effects resulting from the devices.
3. Students are allowed to **use only the official lab PC allocated by the instructor** during the tests. We cannot accept any complaints since the same environment is being used throughout all weeks.
4. **Students must have an active participation in the labs:** an attendant student must use provided PC, open the desired environment, and code together with the instructor. Other activities (such as playing on the PC, talking with a classmate, messaging others, etc.) are considered as unexpected behavior.

5. **Students cannot wear (and use) headphones during the labs and lectures.**
6. In the classroom, **going in and out (for any reason) is not allowed.** Note that this is an inappropriate and unpolite manner that disturbs teaching. In this case, the instructor reserves the right to count you absent.
7. **There is no way to make up a missed class.** The absent students should check the available materials and prepare themselves for the next week. The instructor is not responsible for explaining the skipped lab's material.
8. **Students responsible for their own PCs:** Instructors cannot help to resolve sysadmin problems on the students' personal notebooks.
9. **The official operating system is Windows:** the lab PCs have Windows operating system. Thus, each student should be able to use its basic utilities and the course environment.

Instructors will automatically disqualify a student from the Bonus test offering if he/she breaks the previous guidelines or starts a non-mentioned unethical or suspicious activity (at any time in the semester).

Cheating students on the *Lab test*, *Theory test*, *Lab retake*, or *Theory retake* cannot get their passing grades in this semester.

Check the '*Regulations*' part of the following page:

<https://inf.unideb.hu/en/documents>

<https://edu.unideb.hu/p/rules-and-regulations>

4. Requirements and examination

You can find a comprehensive description of all the requirements and important, must-to-know details about both the labs and lectures. We will consistently use the expressions *Attendance*, *Lab test*, *Retake test*, and *Exam* this semester. The *Midterm* and *Endterm* expressions can be misunderstood in many situations; thus, they will be avoided.

Check frequently the official channels for announcements.

Attendance

1. A student can skip at most three (3) labs. The student has the responsibility to learn missed topics using the lab codes.
2. A student must arrive before the 20th minute of the lab (except the tests, where delay voids your test). In the computer lab, continuously or regularly going in and out (for any reason) is not allowed. Note that this is inappropriate and unpolite behavior. In this case, the instructor reserves the right to count you absent. Instructors can consider a student absent if he/she does not meet the conditions. However, in this case, it is still allowed to stay in the class. There is no way to make up a missed class.
3. The attendance will be registered using a sign sheet. Students must present their IDs to be registered as an attendant.
4. Missed classes are counted from the first week of the semester. There is no transfer of attendance between groups.
5. Students must be presented at the original time of their labs, as they registered in Neptun.
6. Switching between labs is impossible.

Lab test

1. The *Lab test* tests the coding skills and overall knowledge of the student at the end of the semester.
2. Students will get their signatures based on the result of this test.
3. The conditions and details (including the requirements and the grading system) will be announced at least seven (7) days before the test.
4. The grading system can include mandatory criteria that all passing students must fulfill.
5. The mandatory criteria can be evaluated by a script. Additional features will be checked manually by the instructors, giving penalty points for mistakes. The grading sheets will be published prior to the test, and instructors will strictly follow them. Asking for overriding criteria or giving any discounts will be considered an unethical action.
6. A *Lab test (repeat occasion)* is also included in the weekly schedule. Attendance is not managed for those who have already passed.

Retake test

1. Instructors will offer a single retake opportunity for those who failed the *Lab test*.
2. The structure of this test (including the grading system) will be inherited from the *Lab test*.
3. An additional opportunity will be organized in the first exam week.
4. The conditions and details will be announced at least seven (7) days before the test.

5. Test specification

Exercise

1. You will be asked to create a single source code that processes a CSV file.
2. The input must be read from a CSV file; its name (path) will be passed as a command-line argument.
3. The output must be written to a CSV file; its name (path) will be passed as a command-line argument.
4. You must create your own record type (based on a structure) to represent the lines of the document.
5. You must store the records in a suitable array.
6. You must perform a multi-stage ordering on the array using the built-in function `qsort()`.

Students get **40 minutes** (approximately) to implement and upload their submissions using the provided web application (via the clipboard).

Grading

Minimum (mandatory) requirement

A passing solution must fulfill the following requirements:

1. Do not contain a syntax error using the instructor's default MinGW installation using the following command:

```
gcc <your-file-name> -std=c99 -Wall -lm -o solution
```

2. Return exit code `0` when the user specifies all the command-line arguments and all the files can be opened.
3. Implement the *Core functionality* without any restrictions on the source code:
 1. Open a CSV file and read its lines (records).
 2. Sort the records in the memory using the function `qsort()`.
 3. Open a CSV file and write the ordered records.
4. The student must collect at least **70 points**.

Grading criteria

The following table contains the criteria and the corresponding points. It is impossible to give more subpoints; a criterion is fulfilled or not. Instructors may test your source code using another input file that meets the specification; thus, the existence of the core functionality will be decided based on that input.

Criterion	Score
Core functionality. ¹	50
Record - using the <code>typedef</code> keyword	2
Record - name of the type	2
Record - names of the fields	4*2 = 8
Record - types and lengths of the fields	4*1 = 4
Reading - handling situation: no 1st argument	2
Reading - handling situation: file cannot be opened	2
Reading - declaring the record array having a suitable size	2
Reading - declaring the character array having a suitable size	2
Reading - closing the file	1
Sorting - signature of the comparison function	1
Sorting - the body of the comparison function	3
Sorting - invocation of the comparison function	1
Writing - handling situation: no 2nd argument	2
Writing - handling situation: file cannot be opened	2
Writing - closing the file	1
Total	85

Notes

1. Other criteria cannot be evaluated without passing the criterion *Core functionality*.
2. The evaluation of the *Core functionality* criterion is automatized using a `Python` script:
 1. The solution is compiled using the provided `gcc` command.
 2. The solution is executed by providing the **public test case (test file)** (that is the part of the test sheet).

1. The solution passes the test case if the status code is 0 and the output file is identical to the expected one. The comparison is performed character-wise, so the files must be identical.
2. The solution is rejected if the output file differs from the expected one. Each line must be terminated by a newline character.
In this case, the score for the *Core functionality* is 0 (thus, the student fails).
3. The solution is executed by providing the **secret test case (test file)** (that is **NOT** the part of the test sheet).

1. The solution passes the test case if the status code is 0 and the output file is identical to the expected one. The comparison is performed character-wise, so the files must be identical.
2. The solution is rejected if the output file differs from the expected one. Each line must be terminated by a newline character.
In this case, the score for the *Core functionality* is 0 (thus, the student fails).
3. The instructor evaluates the further criteria manually.

6. Contact information

Discussion forms

In this course, we distinguish two categories of questions:

- A question is *technical* if it affects the schedule, the requirements, the attendance, or any other feature of this course.
- A question is *professional* if it is about any lab topic of this course.

Each question should fulfill all the expectations specified in section *Ethical and technical guidelines*. Students can ask questions in the following forms in the following precedence order:

1. **Real-time professional and technical questions:** Your active collaboration and questions are always welcomed (and required) during the labs.

Feel free to raise your hand and ask your instructor about the current professional content at any time during the lab (except for tests). The same applies to all the technical information that are explained by your instructor.

2. **Retrospective technical and professional questions:** You can ask your question online (in Moodle) between two consecutive labs.

Your instructor will read the questions and reflect on them in the first minutes of the next session. These questions can be technical or professional. It is highly recommended to ask your questions about the previous week's content in this form since each lab is based on the previous content. This is why each session's agenda starts with a *Warm-up Q&A session* that will reflect on these questions.

3. **Individual professional questions:** There can be situations (e.g., an instructor cannot detect your bug in the source code during the lab) in which your *instructor will ask you* to visit the office hours or messaging via email.

In these situations, you will be asked to visit the instructor's office hours or to send an email that will be processed before the next lab. The instructor will reply, or the question will be added to the next session's discussion. In this case, please follow the instructions in section *Messages*. Note that this is the option with the least precedence, and your instructor can ask you to use it. Technical questions cannot be classified into this category since the same rules and conditions apply to all the students; thus.

Students must follow all the rules introduced in the section *Ethical and technical guidelines*.

All the possible questions should be public interest and asked as *real-time* or *retrospective* questions. Lab instructors do not provide any information about the lectures. Only questions related to the lab part can be asked.

We do not accept custom emails.

Reviewing the *Lab test* and *Retake test*

Your instructors will guarantee your right to review the tests by publishing the sheets, your solutions, and the uniform grading sheet. This procedure guarantees transparency and equal conditions for all the students. The instructor checks the submissions and provides feedback within five workdays. Students are able to access their submitted codes, the sheets, and the grading sheet online after authentication. They are able to send an inquiry about the result in the next three (3) workdays if the instructor misgraded or miscalculated something. An inquiry should contain a message that describes the miscalculation. The instructor(s) evaluate the inquiries in five (5) workdays and provide feedback on the same platform. Instructors have the right to share and discuss the inquiries with their fellow colleagues using the student's source code (without sharing the student's personal data). The instructor updates the student's grading sheet in the case of an accepted inquiry.

Announcements

General and important announcements from instructors will be sent out using the Neptun and Moodle systems (one or both). It is the student's responsibility to regularly check these channels for messages and announcements, especially before exams and deadlines. *Complaints regarding unseen messages are rejected.*

Messages

Instructors and lecturers do not guarantee that they will answer emails regarding the subject since the mentioned contact forms need to be used as the official communication channels (except if they ask you to message them as *individual professional questions*). Messages with irrelevant topics automatically remain unanswered. (e.g., personal influence about what will happen if exams fail, about scheduled flights and ticket issues, urgent personal problems, etc.) Similar regulations apply to messages sent through LinkedIn, Facebook, Instagram, and other social platforms.

Only your official university-provided (@mailbox.unideb.hu) email address is accepted for email messages. Gmail and similar domains are **not** processed since instructors must identify the student to guarantee privacy about students' sensitive personal information. The expected reply time is three (3) working days. Weekends and holidays are not counted. We recommend and kindly ask you to follow the instructions in the section *Discussion forms* to get a quick response.

Please include the following information in your message:

- Subject for the message
- Date of the issue if relevant
- Name - as Neptun shows
- Neptun code
- Major
- Course title